



Darshan
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Python Programming - 2301CS404

Lab - 8

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}

```
In [1]: a = {1, 2, 3, 4}
        b = {3, 4, 5, 6}

        print(a.union(b)) # or a | b
        print(a.intersection(b)) # or a & b
        print(a.difference(b)) # or a - b
```

```
{1, 2, 3, 4, 5, 6}
{3, 4}
{1, 2}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}

Sample Output:

{2, 3}

```
In [5]: a = {1, 2, 3, 4}
b = {2, 3, 5}
c = {2, 3, 6}

# print((a & b) & c)

# or

print(a.intersection(b.intersection(c)))
```

{2, 3}
{2, 3}

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

[10, 20, 10, 30, 20, 40]

Sample Output:

{10, 20, 30, 40}

```
In [9]: li = [10, 20, 10, 30, 20, 40]

s = set(li)

print(s)
```

{40, 10, 20, 30}

Out[9]: {10, 20, 30, 40}

04) WAP to count how many elements are common between two sets.

Sample Input:

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

```
In [10]: A = {1, 2, 3, 4}
         B = {3, 4, 5, 6}

         print(len(a&b))
```

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05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

```
In [ ]: Dict1 = {1: 10, 2: 20, 3: 30}
         Dict2 = {2: 40, 3: 50, 4: 60}

         for k,v in Dict1.items():
             if k in Dict2:
                 Dict2[k] += v
             else:
                 Dict2[k] = v

         print(Dict2)

         # result = Dict2.copy()

         # for k, v in Dict1.items():
         #     result[k] = result.get(k, 0) + v

         # print(result)
```

```
{2: 60, 3: 80, 4: 60, 1: 10}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

```
In [49]: Dict = {'Math': 70, 'Physics': 85, 'Chemistry': 90}

def insort(item):
    return item[1]

di = sorted(Dict.items(), key=insort)

# or

# di2 = sorted(Dict.items(), key=lambda x : x[1])

# print(dict(di2))

print(dict(di))
```

```
{'Chemistry': 90, 'Physics': 85, 'Math': 70}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

```
In [68]: Dict = {'A': 50, 'B': 90, 'C': 30}

di = sorted(Dict.items(), key=lambda x : x[1])
di = dict(di)

print("max",list(di.keys())[len(di)-1])
print("min",list(di.keys())[0])

# or

# m = max(Dict , key=Dict.get)
# n = min(Dict , key=Dict.get)
```

```
max B
min C
```

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

```
In [71]: Dict = {i : i*i for i in range(1,6,1)}  
print(Dict)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

```
In [ ]: s = "programming"  
  
# di = {}  
  
# for i in s:  
#     di[i] = s.count(i)  
  
di = {i : s.count(i) for i in s}  
  
print(di)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

```
In [76]: di = {1: 'A', 2: 'B', 3: 'C'}  
  
di2 = {j : i for i , j in di.items()}  
  
print(di2)
```

```
{'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

Topper: Amit

```
In [ ]: stu = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}

# for k , v in stu.items():
#     stu[k] = sum(v)

# stu = sorted(stu.items(), key=lambda x : x[1],reverse=True)

stu = dict(sorted(stu.items(),key=lambda x : sum(x[1])))

print("Topper :",stu[0][0])
```

Topper : Amit

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

```
In [85]: di = {1: 11, 2: 20, 3: 15, 4: 40}

li = [k for k,v in di.items() if v % 2 == 0]

print(li)
```

Out[85]: [2, 4]

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

```
In [86]: Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

```
li = []

for i in Dict1.keys():
    if i in Dict2:
        li.append(i)

print(li)
```

Out[86]: [2, 3]

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

8

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

Key Not Found

```
In [ ]: dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'

# if key in dict1:
#     print(dict1[key])
# else:
#     print("Not Found")

dict1.get(key, "Key Not Found")
```

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15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

```
In [92]: students = ['Amit', 'Riya', 'Neha', 'Karan']  
marks = [78, 72, 85, 60]  
  
di = {k:v for k,v in zip(students,marks) if v >= 75}  
  
print(di)
```

```
Out[92]: {'Amit': 78, 'Neha': 85}
```